Appendix C – Study outcomes per included study

Table 1. Study outcomes for included study in the main analysis, where intervention groups were aggregated if there was more than one intervention condition. The means and standard deviations are displayed as reported in each study. These outcomes represent the average diagnostic accuracy in both the comparator group and the intervention group over all cases. Additionally, the interpretation of the results given by the authors of the original study and the source from which the data were obtained are listed.

Study	**Comparator**			**Intervention**			**Effect estimate Hedges' g (95% CI)**	**Interpretation**	**Source**
	N	**Mean**	**SD**	**N**	**Mean**	**SD**			
Berbaum et al. (2006)[1]	19	0.75 ROC[a]	0.87	20	0.67 ROC[a]	0.57	-0.11 (-0.74; 0.52)	"[…] use of a self-prompting checklist to counteract satisfaction of search is not warranted…"	Journal article; SD and effect size were computed from available information

Cairns et al. (2016)[2]	11	45%	18%	20	59%	42%	0.36 (-0.38; 1.11)	"[…] this approach improves diagnostic accuracy at the expense of time…"	Journal article; effect size was calculated from available information
Chartan et al. (2019)[3]	56	48%	50%	53	62%	49%	0.28 (-0.10; 0.66)	"A cognitive strategy of prompting red flag isolation prior to differential diagnosis did not improve diagnostic accuracy of 'do-not-miss diagnoses'."	Journal article; effect size was calculated from available information
Costa Filho et al. (2019)[4]	28	38.4%	14.6%	33	49.7%	12.1%	0.84 (0.31-1.37)	"Deliberate reflection increased diagnostic accuracy in dermatology but did not	Journal article; effect size was calculated from

								affect confidence and calibration."	available information.
Dinardo et al. (2018)[5]	33	52%	44%	33	60%	44%	0.18 (-0.30; 0.66)	"A clinical reasoning tool improved residents' diagnostic accuracy on written cases."	Journal article; SD obtained via correspondence and effect size was calculated from available information
Ely et al. (2015)[6]	7	82%	20%	7	89%	10%	0.39 (-0.67; 1.45)	"Checklists did not improve the diagnostic error rate in this study."	Journal article; effect size was calculated from available information

Graber et al. (2009)[7]	84	40%	49%	33	61%	50%	0.42 (0.02; 0.83)	"In summary, our study indicates that medical trainees use and misuse electronic decision support products in deriving a differential diagnosis for a challenging cases."	Journal article; effect size was calculated from available information
Ilgen et al. (2011)[8]	115	1.73[b]	0.49	115	1.97[b]	0.47	0.49 (0.22; 0.75)	"Overall, mean diagnostic accuracy and the extent to which the test consistently discriminated between subjects was higher when participants were given directed search instructions than when they were given first impression instructions."	Journal article; effect size was calculated from available information

Ilgen et al. (2013)[9]	192	63.8%	17.3%	201	61.1%	17.4%	-0.16 (-0.35; 0.04)	"Instructions to trust one's first impressions result in similar performance when compared with instructions to consider clinical information in a systematic fashion, but have greater utility when used for the purposes of assessment."	Journal article; effect size was calculated from available information.
Kämmer et al. (2021)[10]	30	0.49[c]	0.50	60	0.57[c]	0.48	0.16 (-0.28; 0.60)	"The use of DDXC but not of GDBC may enhance diagnostic accuracy, without influencing the diagnostic reasoning process, but only if it contains the correct diagnosis."	Journal article, effect size was calculated from available information

Kilian et al. (2019)[11]	17	53%	50%	17	57%	50%	0.07 (-0.60; 0.74)	"Decisions to revise diagnoses may be cued by the detection of contradictory evidence."	Journal article; effect size was calculated from available information
Kok et al. (2017)[12]	20	33.13%	25.87%	20	38.37%	24.93%	0.20 (-0.42; 0.82)	"In conclusion, a checklist can help medical students to detect abnormalities in chest radiographs."	Journal article; effect size was calculated from available information
Lambe et al. (2018)[13]	54	0.47 [c]	0.26	112	0.42 [c]	0.25	-0.22 (-0.54; 0.11)	"This study finds no evidence to support the use of the guided reflection method as a diagnostic aid for novice diagnosticians…"	Journal article; effect size was calculated from available information

Li et al. (2020)[14]	46	0.67 [c]	0.45	46	0.76 [c]	0.40	0.21 (-0.20; 0.62)	"Availability bias led to diagnostic errors. Misdiagnoses cannot always be repaired by solely adopting a reflective approach."	Journal article; effect size was calculated from available information
Mamede et al. (2008)[15]	42	0.38 [c]	0.21	42	0.40 [c]	0.23	0.09 (-0.34; 0.52)	"Reflective practice had a positive effect on diagnosis of complex, unusual cases. Non-analytical reasoning was shown to be as effective as reflective reasoning for diagnosing routine clinical cases."	Journal article; effect size was calculated from available information
Mamede et al. (2010a)[16]	84	0.49 [c]	0.34	84	0.54 [c]	0.29	0.16 (-0.15; 0.46)	"Experts benefit from consciously thinking about complex problems; for novices	Journal article; raw data were obtained via correspondence

								thinking does not help in those cases."	and effect sizes were calculated from available information
Mamede et al. (2010b)[17]	36	0.23[b]	0.17	36	2.17[b]	1.04	0.38 (-0.09; 0.84)	"When faced with cases similar to previous ones and using nonanalytic reasoning, second-year residents made errors consistent with the availability bias. Subsequent application of diagnostic reflection tended to counter this bias; it improved diagnostic accuracy in both first- and second-year residents."	Journal article; effect size was calculated from available information

Mamede et al. (2020)[18]	45	0.23 [c]	0.17	122	0.27 [c]	0.17	0.25 (-0.09; 0.60)	"Physicians' diagnostic accuracy improved after reflecting upon initial diagnoses provided for difficult cases, independently of the evidence searched for while reflecting."	Journal article; effect size was calculated from available information
Martinez-Franco et al. (2018)[19]	44	74.1%	9.4%	43	82.4%	8.5%	0.92 (0.47; 1.36)	"Family Medicine residents have appropriate diagnostic accuracy that can improve with the use of DXplain. This could help decrease diagnostic errors, improve patient safety and the quality of medical practice."	Journal article; effect size was calculated from available information

Myung et al. (2013)[20]	80	3.05[b]	0.98	65	3.40[b]	0.66	0.41 (0.08; 0.74)	"Enhancement of analytic reasoning may improve diagnostic accuracy in novice doctors."	Journal article; effect size was calculated from available information
Nickerson et al. (2020)[21]	50	6.80[b]	1.60	50	7.20[b]	1.40	0.26 (-0.13; 0.66)	"Using a checklist with common syncope-related pathology when interpreting an ECG for a patient with clinical scenario of syncope may improve residents' ability to recognize some clinically important pathologies; however it could lead to increased interpretation and	Journal article; effect size was calculated from available information

								suspicion of pathology that is not present."	
O'Sullivan et al. (2019)[22]	37	3.10[b]	1.86	37	2.80[b]	1.86	-0.16 (-0.62; 0.30)	"There is insufficient evidence to recommend this tool in clinical practice, however the qualitative data suggests such an approach has some merit and face validity to users."	Journal article; SD and effect size were calculated from available information
Schmidt et al. (2014)[23]	38	0.56[c]	0.23	38	0.71[c]	0.3	0.56 (0.10; 1.01)	"Availability bias may arise simply from exposure to media-provided information about a disease, causing diagnostic errors. The bias's effect can be substantial. It is apparently associated with	Journal article; effect size was calculated from available information

								nonanalytical reasoning and can be counteracted by reflection."	
Schmidt et al. (2017)[24]	63	0.59[c]	0.29	63	0.64[c]	0.29	0.17 (-0.18; 0.52)	"Disruptive behaviours displayed by patients seem to induce doctors to make diagnostic errors."	Journal article; effect size was calculated from available information
Shimizu et al. (2013)[25]	188	1.55[d]	1.04	188	1.67[d]	1.14	0.11 (-0.09; 0.31)	"The use of DDXC, not GDBC, may improve the diagnostic performance in difficult cases, while intuitive process may still be better for simpler cases."	Journal article; raw data were obtained via correspondence; SD and effect size were calculated

									from available information
Sibbald et al. (2013)[26]	191	46%	50%	191	51%	50%	0.09 (-0.11; 0.29)	"Verifying diagnostic decisions with a checklist improved diagnostic accuracy. This benefit was only seen when more information could be collected."	Journal article; effect size was calculated from available data
Staal et al. (Unpublished)	42	55%	20%	39	62%	20%	0.35 (-0.09; 0.31)	"Checklist use improved confidence-accuracy calibration in normal as well as abnormal cases."	Raw data obtained from authors
Talebian et al. (2014)[27]	31	42%	49%	31	76%	43%	0.74 (0.50; 0.97)	"The increased accuracy of ECG interpretation and final diagnosis can be attributed to	Journal article; raw data were obtained via

								the utilization of a checklist by residents, especially in the first year and second year residents."	correspondence; SD and effect size were calculated from available data
Thompson et al. (2017)[28]	14	77%	5%	14	92%	5%	2.96 (1.84; 4.08)	"This study demonstrates students can quickly and effectively learn to read a lateral chest film using this novel mnemonic."	Journal article; effect size was calculated from available data

[a] ROC: receiving operator characteristic.

[b] Average number of cases correct (range: 0-4 ([16 20]), 0-10 ([21 22]), 0-12 ([8])).

[c] Each case was scored as incorrect (0), partially correct (0.5), or correct (1). These scores were averaged over all cases.

[d] A score of 3 was given if the correct diagnosis was the first most likely diagnosis, a score of 2 if the correct diagnosis was the second most likely diagnosis, a score of 1 if the correct diagnosis was the third most likely diagnosis, and 0 if the diagnosis was not included. These scores were averaged over all cases.

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